Unit XIII: Aldehydes and Ketones

Video lecture notes: what she explained and wrote on the slides

CHEMISTRY (PROF. L. RANAY)

WHERE THIS FITS

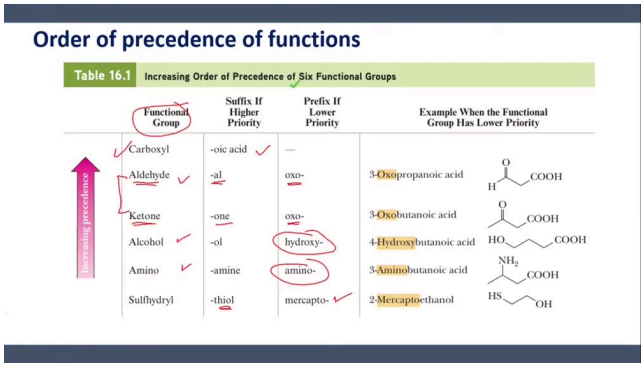
- Second-to-last family chapter: the **carbonyl** group (C=O) families. Same outline as before: structure, naming, physical properties, chemical properties.
- This chapter: aldehydes and ketones. Next chapter (Unit XIV): carboxylic acids and their derivatives.

STRUCTURE

- **Aldehyde:** carbonyl carbon attached to **at least one H**. Simplest = one carbon, with H on both sides (methanal).
- **Ketone:** carbonyl carbon attached to **two other carbons**. Smallest ketone = **three carbons: propanone (acetone)**. With only H on one side it is already an aldehyde, not a ketone.
- An aldehyde and a ketone with the same number of carbons are **functional group isomers** (same molecular formula).
  - Example she worked out: propanone and propanal are both **C_3H_6O**.
- The only difference: in an aldehyde the carbonyl carbon holds an H.

ORDER OF PRECEDENCE (NAMING PRIORITY)

- Highest to lowest: **carboxyl (-oic acid) > aldehyde (-al; prefix oxo-) > ketone (-one; prefix oxo-) > alcohol (-ol; prefix hydroxy-) > amine (-amine; prefix amino-) > thiol (-thiol; prefix mercapto-)**.
- Carboxyl is the top priority, so it is never a branch, except in very complicated structures.
- A molecule with both: aldehyde wins over ketone, so the parent name is the aldehyde.
- When OH, NH_2 or SH is present with a carbonyl, it becomes the prefix hydroxy-, amino- or mercapto-.

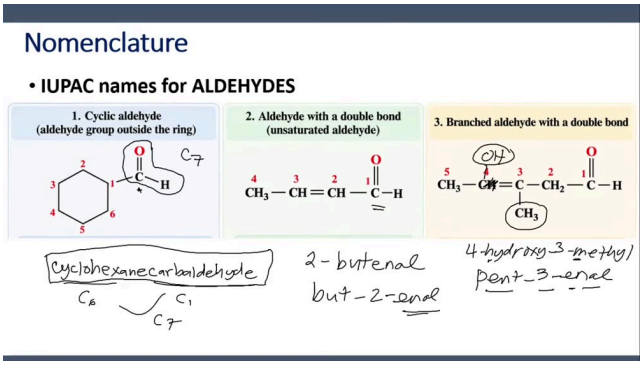


Her marks on the order-of-precedence table: circled prefixes hydroxy- and amino-, ticks down the list

NAMING ALDEHYDES

- Same rule: longest continuous chain that holds the functional group. Drop the **-e**, add **-al**.
- The CHO carbon is **always C1 and always at the end**, so **no locator** for the aldehyde.
- Her board examples (she circled the carbonyl, the OH and the numbering, and wrote the names under the structures):
  - CH_3CH_2CHO: **propanal**.
  - Branched: **3-methylbutanal**. Number from the CHO carbon.
  - With an OH lower in priority: **4-hydroxybutanal**.
- **Cyclic aldehyde: the CHO can NEVER be inside a ring.**
  - Her proof: a ring carbon already has two ring bonds; a C=O plus an H would give **five bonds**, impossible. Carbon holds only four.
  - So the CHO is always outside the ring. Name the ring, then add **carbaldehyde**: **cyclohexanecarbaldehyde**.
  - It is **C7** in total: cyclohexane is C6, the carbaldehyde carbon is C1.
- **Double bond:** the C=C stays in the parent chain, but the aldehyde still decides C1. Example: **but-2-enal** (also written 2-butenal). The "-en" and "-al" are written together.
- Her harder board example: a five-carbon aldehyde with a C=C, a methyl and an OH: **4-hydroxy-3-methylpent-3-enal**.

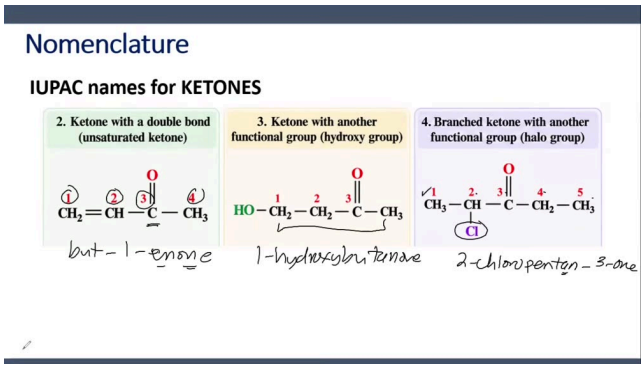
- Substituents are listed **alphabetically** (hydroxy before methyl), then the number of the double bond, then the functional group.



Her board work for aldehydes: cyclohexanecarbaldehyde (C6 + C1 = C7), but-2-enal, 4-hydroxy-3-methylpent-3-enal

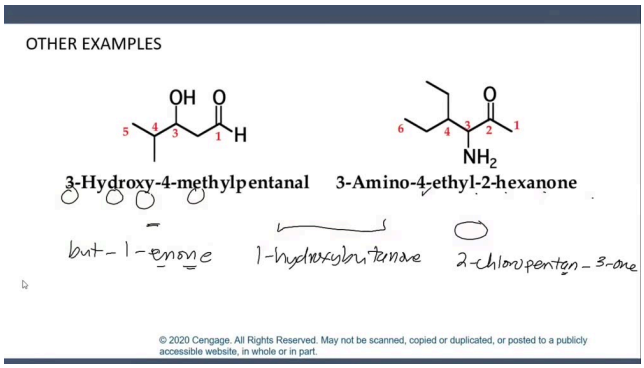
NAMING KETONES

- C=O sits **between carbons**. Drop -e, add **-one**.
- **Propanone** and **butanone** need **no locator**: there is only one possible position for the C=O (there is no 3-propanone and no 3-butanone).
- **From five carbons up you must locate it**: pentan-2-one and pentan-3-one are different compounds (older style: 2-pentanone and 3-pentanone). She showed both on the board.
- **Cyclic ketones:** the carbonyl carbon is **part of the ring**, unlike aldehydes. Cyclohexanone, cyclopentanone. No number needed: it is always carbon 1 in a ring.
- Numbering order (she repeated it): functional group first, then the double bond, then the nearest branch, then alphabetical.
  - **Highest priority is the carbonyl; the next priority is the branch.** Give the branch the lowest number you can.
- 2-chloropentan-3-one: five carbons, so the C=O is located (3), then the chloro is on C2.
- Other examples she did: **3-hydroxy-4-methylpentanal** (aldehyde, no locator) and **3-amino-4-ethylhexan-2-one** (older style: 3-amino-4-ethyl-2-hexanone). "Hexan-2-one" is the **latest IUPAC** style.



Her board work for ketones: propanone, butanone, pentan-2-one, pentan-3-one, cyclohexanone, cyclopentanone, 2-chloropentan-3-one

Note: Her butanone shortcuts. On the board she named $CH_3–CH(CH_3)–CO–CH_3$ as "**2-methylbutanone**" (and marked "3-methylbutanone" wrong), and wrote "1-hydroxybutanone" and "but-1-enone" for butanones with an OH or a C=C. These follow her rule "no locator for butanone, so number to give the branch the lowest number". Current IUPAC writes **3-methylbutan-2-one**, **4-hydroxybutan-2-one** and **but-3-en-2-one**. The question bank does NOT quiz these disputed names. Learn her rule, but if you are asked to write a butanone with a branch, follow what she wrote in class.



Other examples: 3-hydroxy-4-methylpentanal (aldehyde circled) and 3-amino-4-ethylhexan-2-one (she wrote the newer style by hand)

PHYSICAL PROPERTIES

- Both families contain oxygen, so the **C=O is polar** (O is highly electronegative: δ− on O, δ+ on C). Both are polar molecules.
- But only **simple polar**: no H on the oxygen, so **no hydrogen bonding** between their own molecules. The intermolecular force is **dipole-dipole**.
- So they boil **lower than alcohols and carboxylic acids** (which hydrogen-bond).

- **Boiling points at similar molecular weight** (table she walked through): diethyl ether 34 < pentane 36 < butanal 76 < 2-butanone 80 < 1-butanol 117 < propanoic acid 141 °C.
  - Ether is lowest: only slightly polar.
  - Pentane (straight chain) is a little higher than the ether.
  - Butanal and butanone have a **C=O**, which is more polar than an ether, so they boil higher.
  - Butanone is slightly higher than butanal (she said some books treat them as almost equal).
  - **More branching = lower boiling and melting point** (packing).
- **Solubility:** dipole-dipole, so only some solubility. Smaller ones (formaldehyde, acetaldehyde, acetone) dissolve in water infinitely. Solubility **falls as the carbon part grows**. Comparing with ethers (same type: polar, no H bonding) she gave her boundary: **C3 slightly soluble, C4 insoluble** (confirmed in both the Whisper and the YouTube transcripts).

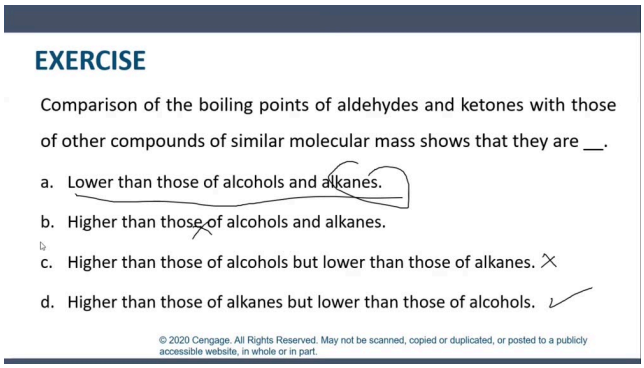
Note: The slide lists formaldehyde, acetaldehyde and acetone as infinitely soluble; her spoken boundary (C3 slightly, C4 insoluble) is the one she gave for ranking, the same as her ether numbers.

- **Exercise (answered in class): aldehydes and ketones boil higher than alkanes but lower than alcohols → letter D.**

OXIDATION AND REDUCTION (THE KEY DIFFERENCE BETWEEN THE TWO)

- **Aldehydes can be oxidized (to carboxylic acids); ketones resist oxidation.** This is how you tell them apart.
- Review she gave:
  - 1° alcohol → oxidized to **aldehyde** → oxidized again to **carboxylic acid**.
  - 2° alcohol → oxidized to **ketone** → **no further reaction**.
  - 3° alcohol → **cannot be oxidized**.
- **Reduction is the reverse** of oxidation: forward = oxidation, backward = reduction.
  - **Aldehyde → primary alcohol** (CH_2OH).
  - **Ketone → secondary alcohol** (the carbon keeps one H and two carbons).
  - Her board sketch: reduced aldehyde carbon has two H's and one carbon (primary); reduced ketone carbon has one H and two carbons (secondary).
- Oxidizing agents she named and wrote above the arrow: **$K_2Cr_2O_7$** (she said "potassium chromate"; circled on the slide), **$KMnO_4$** and **PCC**.
- She wrote **alcohol dehydrogenase** on the slide: the enzyme that does this oxidation in the body.
- Her summary on the oxidation slide, in her own shorthand:
  - **1° alcohol → [O] → aldehyde (RCHO) → [O] → RCOOH.**
  - **2° alcohol → [O] → ketone (RCOR') → [O] ✗ NR (no reaction).**
  - **3° alcohol → [O] ✗ NR.**

Note: Reason for aldehyde oxidation. She said it is "due to the acidic alpha hydrogen" that aldehydes have and ketones lack (both transcripts agree on those words). What actually matters: an aldehyde has an **H on the carbonyl carbon** (a ketone has none). The bank uses that wording.



EXERCISE

Comparison of the boiling points of aldehydes and ketones with those of other compounds of similar molecular mass shows that they are __.

a. Lower than those of alcohols and alkanes.

b. Higher than those of alcohols and alkanes.

c. Higher than those of alcohols but lower than those of alkanes. ✗

d. Higher than those of alkanes but lower than those of alcohols. ✓

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Her oxidation summary written on the slide: 1° to RCHO to RCOOH, 2° to RCOR' then no reaction, 3° no reaction; KMnO4, PCC, K2Cr2O7; alcohol dehydrogenase

TESTS THAT TELL ALDEHYDE FROM KETONE

- **Tollens' test:** ammoniacal silver nitrate ($Ag(NH_3)_2^+$). An aldehyde is oxidized and **silver deposits as a silver mirror** on the test tube. A ketone gives no mirror.
- **Benedict's test** (she also mentioned Fehling's): sugars are aldehydes, so with copper sulfate a **brick-red precipitate** shows sugar in urine. No sugar = no brick-red precipitate.

REDUCTION REAGENTS

- Lab: **H_2 with a nickel catalyst** (exercise below).
- In the body: **NADH** (nicotinamide adenine dinucleotide, reduced form). NADH is the

reducing agent and becomes NAD^+ when it oxidizes.
  - **Pyruvate (a ketone) is reduced to lactate** when the body lacks oxygen. Lactate builds up during strenuous exercise.
- Not discussed aloud: $NaBH_4$ and the hydride mechanism on the slides.

EXERCISES ANSWERED IN CLASS

What is the product when propanone is reduced with H_2 gas and a Ni catalyst?

2-propanol (a 2° alcohol). Reduction forms an alcohol; choice (a). The C=C "enols" among the choices are not alcohol products.

Which compound is resistant to oxidation with a mild oxidizing agent?

The ketone (propanone). The aldehyde and the 1° alcohol can be oxidized, so answer **(c)**.

Table 16.1 Boiling Points for Six Compounds of Comparable Molecular Weight.

Name	Structural Formula	Molecular Weight (amu)	Boiling Point (°C)
diethyl ether	$CH_3CH_2OCH_2CH_3$	74	34
pentane	$CH_3CH_2CH_2CH_2CH_3$	72	36
butanal	$CH_3CH_2CH_2CHO$	72	76
2-butanone	$CH_3CH_2COCH_3$	72	80
1-butanol	$CH_3CH_2CH_2CH_2OH$	74	117
propanoic acid	CH_3CH_2COOH	74	141

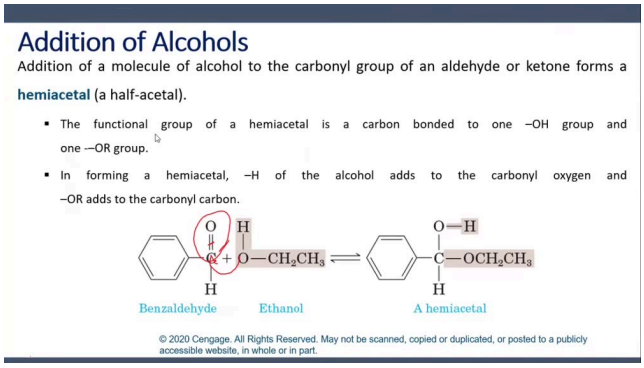
- Formaldehyde, acetaldehyde, and acetone are infinitely soluble in water.
- Aldehydes and ketones become less soluble in water as the hydrocarbon portion of the molecule increases in size.

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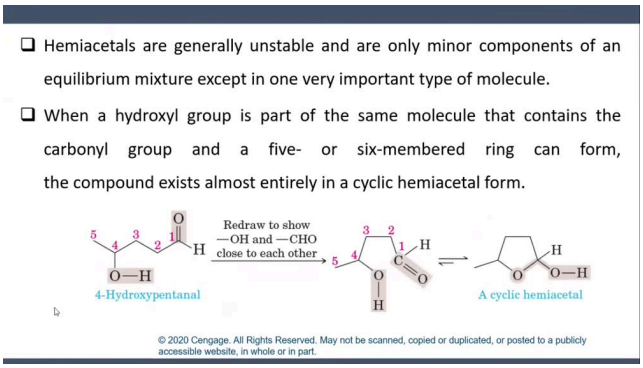
Boiling-point exercise: she marked (d) as the answer

ADDITION OF ALCOHOLS: HEMIACETALS AND ACETALS

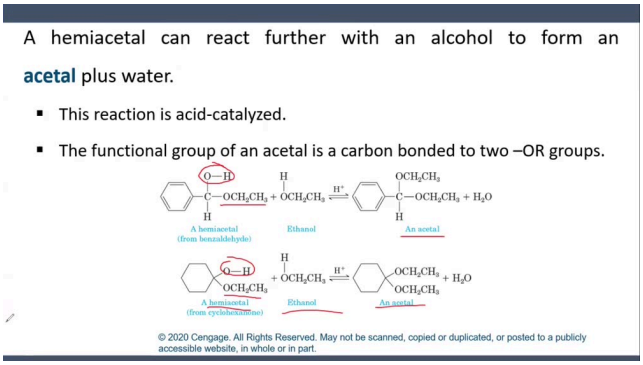
- The carbonyl has a **π bond**, so it can undergo an **addition reaction**. Here the added molecule is an **alcohol**.
- **Both aldehydes and ketones** do it (in an aldehyde the other group is H, in a ketone it is a carbon).
- **Hemiacetal:** one alcohol adds. The **H goes to the carbonyl O**; the **OR (alkoxy) goes to the carbonyl carbon**. The carbon is δ+, the oxygen of the alcohol is the electron pair donor (δ−). The π bond is gone.
  - Product: the carbon now holds **one –OH and one –OR**. "Hemi" = only half.
  - Her example: **benzaldehyde + ethanol → hemiacetal**.
- **Acetal:** a **second alcohol** adds; **the OH is replaced by OR and water is lost** (the H and the OH form H_2O). The carbon now holds **two –OR groups**.
  - Her example: the benzaldehyde hemiacetal + another ethanol → **acetal + water**.
- **What you need:** an aldehyde or ketone **plus an alcohol**.
  - If the other reagent is $KMnO_4$ (an oxidizing agent), it is **oxidation**, not hemiacetal formation.
- **Cyclic hemiacetal:** when the OH and the C=O are in the same molecule, the OH oxygen bonds to the carbonyl carbon and a ring forms.
  - Slide exercise she drew: **5-hydroxyhexanal** (OH on C5, CHO on C1). The OH oxygen bonds to C1 and closes a **six-membered ring** (5 carbons + the O). The old C=O carbon ends up holding the new **–OH and the ring O** (the hemiacetal carbon); the CH_3 stays on the ring carbon next to the O. Her red sketch showed the numbering 1 to 6 and the H leaving the OH.
  - She also drew the open-chain reaction beside it: **aldehyde + alcohol → hemiacetal**.
  - Slide: a five- (4-hydroxypentanal) or six-membered ring forms almost completely. Open-chain hemiacetals are unstable.
- Her trick for exercises: look at the C=O carbon and its oxygen. The oxygen receives the H, the O of the alcohol bonds to the carbon. If either group on the carbonyl carbon is H, it is an aldehyde-derived hemiacetal/acetal.



Her hemiacetal drawing: the carbonyl circled, H to the oxygen, OR to the carbon, and her acetal sketch with + H2O in the corner



The acetal slide with the OH and OR groups circled



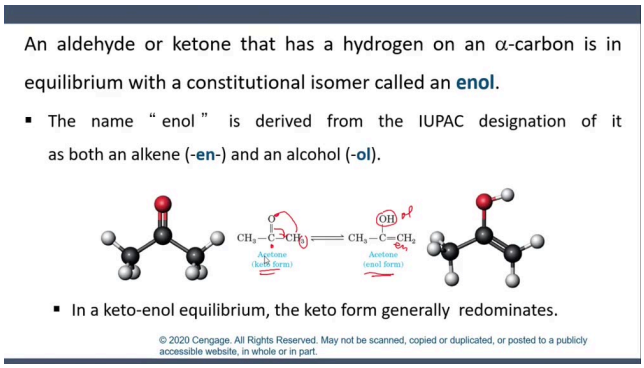
Slide exercise: 5-hydroxyhexanal closes into a six-membered cyclic hemiacetal. Her sketch: the OH oxygen bonds to the carbonyl carbon

KETO-ENOL TAUTOMERISM

- A kind of isomerism: the same substance exists as two isomers that interconvert. Left standing, it is sometimes enol, sometimes ketone.
- Needs a **C=O with an α-carbon that has an α-hydrogen**. The carbon next to the C=O is the **α-carbon**; then **β**, then **γ**.
- The **H moves to the O, the C=O π bond shifts to become a C=C**: ketone (keto form) ⇌ **enol** (alkene + alcohol, "-en" + "-ol").
- The **keto form is the more stable one**. She gave "about 70% keto, 30% enol" in class.

> **Note:** Her 70/30 figures are not accurate for simple ketones (the keto form is far more than 70%). Only the idea "keto form predominates" is tested.

- Shows that carbonyl compounds can form **resonance-like structures**.
- **Exercise (answered in class): draw the two enol forms of each ketone.** In both (a) 2-methylcyclohexanone and (b) pentan-2-one, the new C=C can form on **either side of the C=O**, so each ketone has **two enol forms**: the C=C goes to one α-carbon or to the other.



Her two enol forms for each ketone: the C=C can form on either side of the C=O

BIOLOGICAL KETONES

- The steroid hormones **cortisol** and **aldosterone** (slide exercise): she pointed to the **ketone** in cortisol and said aldosterone has an **aldehyde together with a ketone**. Our hormones carry many carbonyl (ketone) groups.

WHAT TO REVIEW

- Order of precedence, naming (no locator for aldehyde / propanone / butanone; carbaldehyde on a ring), boiling-point ranking, oxidation vs reduction products, Tollens and Benedict tests, hemiacetal vs acetal and what is needed, keto-enol forms.

END OF LESSON